Section A

Q-1 What factors should be considered when selecting the CPU, RAM, and storage to ensure good performance and hardware compatibility?

Factors to Consider When Selecting CPU, RAM, and Storage

To ensure good performance and hardware compatibility at Nexus Innovations, the following factors should be considered:

1. CPU (Processor)

- Cores and threads: More cores and threads help run programming software, browsers, and video conferencing simultaneously.
- Clock speed: A higher clock speed improves performance for demanding applications.
- Compatibility: The CPU must be compatible with the motherboard's socket, chipset, and BIOS.

2. RAM (Memory)

- Capacity: At least 16 GB of RAM is recommended for multitasking; 32 GB is better for heavier workloads.
- Speed: RAM speed should be supported by the motherboard and CPU.
- Compatibility: Check the RAM type (DDR4 or DDR5), motherboard support, and available memory slots.

3. Storage

- Type: An SSD is recommended because it provides faster boot times and application loading than an HDD.
- Capacity: Choose enough storage for the operating system, programming tools, applications, and files.
- Compatibility: Ensure the motherboard supports the storage interface, such as SATA or M.2 NVMe.

q-2 Explain how the antivirus software affects CPU, RAM, and storage performance, leading to system slowdowns.

Antivirus software protects computers from viruses, malware, and other security threats. However, it can slow down system performance by using hardware resources.

1. CPU (Processor)

- Antivirus software uses CPU power to scan files, detect threats, and monitor programs in real time.
- Full system scans can increase CPU usage, making other applications run slowly.

2. RAM (Memory)

- Antivirus software uses RAM to run background services and store scanning data.

- If RAM usage becomes too high, fewer resources remain available for other applications, reducing multitasking performance.

3. Storage (HDD/SSD)

- Antivirus scans read many files from storage, increasing disk activity.
- On an HDD, frequent file access can slow down other operations. On an SSD, heavy scanning can also increase storage activity and cause delays.

Solutions:

- Schedule full scans during non-working hours.
- Keep antivirus software updated.
- Disable unnecessary startup applications.
- Upgrade RAM if memory is insufficient.

q-3 Describe the steps you would follow to troubleshoot the problem if it is caused by

driver incompatibility or operating system issues

Troubleshooting Driver Incompatibility or Operating System Issues

If a USB keyboard and printer work on one computer but not another, I would follow these troubleshooting steps:

1. Check USB Connections: Connect the keyboard and printer to different USB ports. Test them on another computer to confirm that the devices are working correctly.

2. Check Device Manager: Open Device Manager in Windows and look for devices with a yellow warning symbol or error message.

3. Update or Reinstall Drivers: Download the correct drivers from the manufacturer's official website. Update or uninstall and reinstall the incompatible drivers.

4. Run Windows Troubleshooter: Use the Windows hardware or printer troubleshooter to identify and fix common device problems.

5. Check Operating System Updates: Install pending Windows updates and ensure the operating system supports the connected devices.

6. Restart the Computer: Restart the system after installing drivers or updates, then reconnect the devices and test them.

7. Check System Files: If the problem continues, run sfc /scannow in Command Prompt as an administrator to check and repair corrupted Windows system files.

q-4Explain how a UPS and surge protector help protect computer hardware and prevent

data loss during these situations.

Nexus Innovations experiences frequent power cuts and voltage fluctuations. A UPS and a surge protector help protect computer hardware and prevent data loss.

1. UPS (Uninterruptible Power Supply)

- Backup power: Provides temporary battery power during a power cut, allowing employees to save their work and shut down computers safely.
- Voltage regulation: Helps manage voltage fluctuations and provides stable power, depending on the UPS model.
- Data protection: Prevents sudden shutdowns that may cause data loss or file corruption.

2. Surge Protector

- Voltage spike protection: Protects computers, monitors, and other connected devices from sudden voltage spikes.
- Hardware protection: Reduces the risk of damage to sensitive components such as the motherboard, CPU, and power supply.
- Electrical safety: Diverts excess transient voltage away from connected equipment.

q-5 Explain how DNS and DHCP work together in this network, and identify which service is most likely misconfigured based on the symptoms described

How DNS and DHCP Work Together

1. DNS (Domain Name System)

- DNS translates domain names or hostnames into IP addresses.
- It allows employees to access internal servers using names instead of remembering IP addresses.
- If DNS is misconfigured, employees may reach the application server by IP address but not by name.

2. DHCP (Dynamic Host Configuration Protocol)

- DHCP automatically assigns IP addresses to computers and other network devices.
- It also provides the subnet mask, default gateway, and DNS server address.
- If DHCP is misconfigured, devices may fail to obtain a valid IP address when connecting to the network.

3. How DNS and DHCP Work Together

DHCP provides devices with their network configuration, including the DNS server address. DNS then resolves server names into IP addresses, allowing employees to access internal systems by name.

4. Most Likely Misconfigured Services

- DNS: Likely misconfigured because employees can access the application server by IP address but not by name.
- DHCP: May also be misconfigured because some devices fail to obtain valid IP addresses.

Conclusion: Both DNS and DHCP should be checked. DNS is the likely cause of the name-resolution problem, while DHCP may be responsible for the IP-address assignment problem. The IT team should verify DNS records, DHCP scopes, and the DNS server address distributed by DHCP.

q-6 Describe the troubleshooting steps you would take to determine whether the firewall rules are blocking legitimate traffic, and how you would resolve the issue.

Troubleshooting Firewall Rules Blocking Legitimate Traffic

If employees can access the internet but cannot reach the internal application server after a new firewall is installed, I would follow these steps:

1. Check Network Connectivity: Ping the application server's IP address to determine whether it is reachable. Verify the server IP address and network configuration.

2. Check Firewall Rules: Review the firewall rules to see whether legitimate traffic to the internal server is being blocked.

3. Check Ports and Protocols: Verify that the required ports and protocols, such as TCP port 443 for HTTPS or TCP port 80 for HTTP, are allowed.

4. Review Firewall Logs: Examine firewall logs for dropped or denied connections between employee computers and the application server.

5. Test Connectivity: Use commands such as ping, tracert (Windows), or traceroute (Linux) to help identify where connectivity fails. Use Test-NetConnection on Windows to test a specific TCP port.

6. Correct Firewall Rules: Add or modify a rule to allow legitimate traffic from the authorized internal network to the application server on the required ports. Keep the rule as restrictive as possible.

7. Retest and Monitor: Test access to the application server and review firewall logs to confirm that the issue is resolved without allowing unauthorized traffic.

Section – B

Task 1: Workstation Assembly

Step 1: Prepare the Work Area

- Switch off and unplug the power supply.
- Keep the motherboard and components on an anti-static surface.
- Wear an anti-static wrist strap to prevent electrostatic discharge (ESD).

Step 2: Install the CPU

- Open the CPU socket on the motherboard.
- Align the CPU with the socket marking and place it carefully.
- Lock the socket according to the manufacturer's instructions.

Step 3: Install the CPU Cooler

- Apply thermal paste if it is not pre-applied.
- Attach the CPU cooler securely.
- Connect the cooler fan cable to the CPU_FAN header.

Step 4: Install RAM

- Open the RAM slot clips.
- Align the notch on the RAM module with the slot.
- Press gently until the clips lock into place.

Step 5: Install Storage

- Install the M.2 SSD into its slot or mount a SATA SSD/HDD in the drive bay.
- Secure the drive properly.
- For SATA drives, connect the SATA data cable to the motherboard and the SATA power cable to the PSU.

Step 6: Install the Motherboard

- Fit the I/O shield if it is separate.
- Install the motherboard standoffs in the correct positions.
- Place the motherboard inside the chassis and secure it with screws.

Step 7: Install the Power Supply (PSU)

- Position the PSU in its designated compartment.
- Secure it with screws.
- Ensure its cables can reach the motherboard and other components.

Step 8: Connect Internal Cables

- Connect the 24-pin ATX motherboard power cable.
- Connect the 4-pin or 8-pin CPU power cable.
- Connect storage power and data cables where required.
- Connect the front-panel power switch, reset switch, and LED connectors according to the motherboard manual.
- Connect front USB and audio cables.
- Connect case fans to the appropriate headers.

Step 9: Install Additional Components

- Install the graphics card if required.
- Secure it to the chassis and connect the necessary PCIe power cables.
- Check that all components are firmly installed.

Step 10: Perform Final Checks

- Check that all cables are connected correctly.
- Ensure no loose screws or cables obstruct the fans.
- Close the chassis and connect the monitor, keyboard, and power cable.

Step 11: Power On and Verify POST

- Switch on the PSU and press the computer's power button.
- Check that the fans operate and the display shows the startup screen.
- Enter the BIOS/UEFI and verify that the CPU, RAM, and storage are detected.
- Confirm that no POST errors or warning beeps occur.

The workstation assembly procedure is complete when the computer powers on, passes POST without errors, and correctly detects the CPU, RAM, and storage in BIOS/UEFI.

Task 2: RAM Mismatch Troubleshooting

Step-by-Step Procedure

Step 1: Identify the RAM Mismatch Error

- Switch on the workstation and observe the POST messages on the screen.
- If a memory error appears, such as Memory Configuration Error or Memory Size Changed, note the message.
- Enter BIOS/UEFI by pressing the appropriate key, such as Del, F2, or another key specified by the manufacturer.
- Check the detected RAM capacity, speed, and installed memory modules.

Step 2: Determine the Root Cause

Check for the following possible causes:

- Mismatched speed: Installed RAM modules have different rated speeds.
- Mismatched capacity: RAM modules have different capacities, or a module is not detected.
- Incompatible timings: Memory timing settings are unsupported by one or more modules.
- Improper seating: A RAM module is not fully inserted into its slot.

Step 3: Shut Down and Inspect the RAM

- Shut down the computer and disconnect the power cable.
- Follow ESD precautions before touching internal components.
- Remove the RAM modules carefully.
- Check the modules for damage and verify their specifications.
- Reseat the RAM modules firmly in the motherboard's recommended slots.

Step 4: Reconfigure BIOS/UEFI Settings

- Restart the computer and enter BIOS/UEFI.
- Load default memory settings if incorrect manual settings are suspected.
- Disable XMP/EXPO temporarily if an overclocked memory profile is causing instability.
- Allow the system to use a supported memory speed and automatic timings.
- Save the settings and restart the workstation.

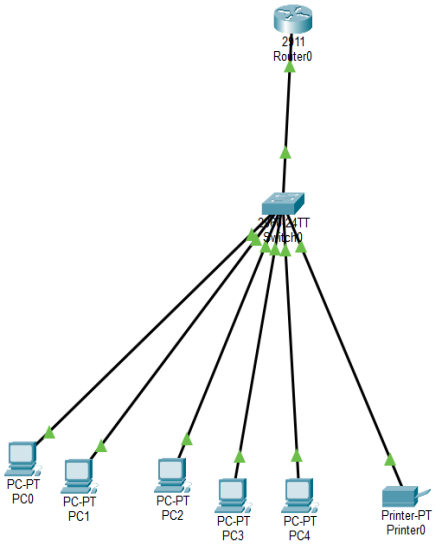
Step 5: Verify the Fix

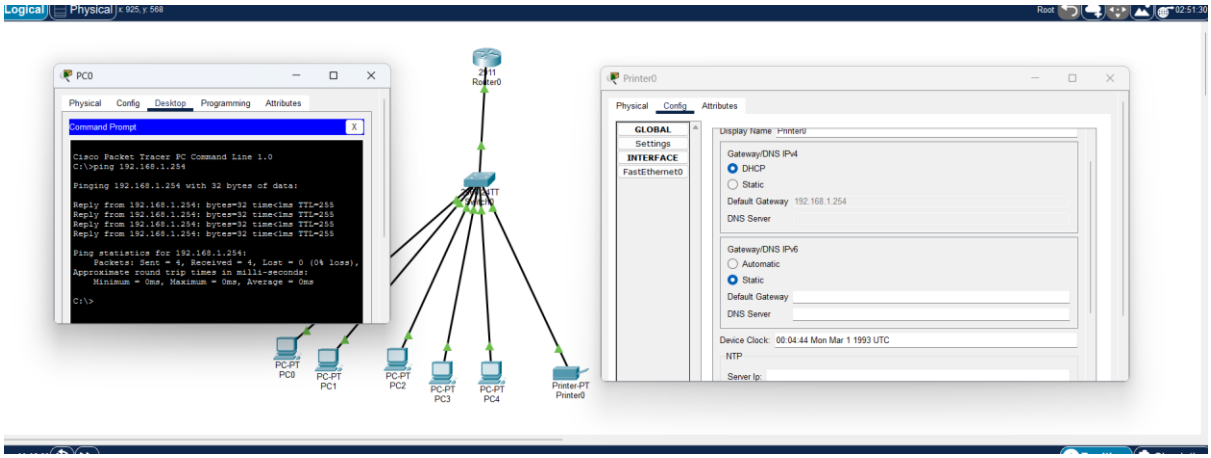
- Check whether the POST error has disappeared.
- Confirm that BIOS/UEFI detects the correct RAM capacity and a supported speed.
- Allow the operating system to boot normally.
- Run Windows Memory Diagnostic or another trusted memory test to check for errors.

Result

The RAM mismatch issue is resolved when the workstation passes POST without memory errors, detects the installed RAM correctly, and boots into the operating system normally.

Task 3: LAN Topology Build





Step 1: Build the Network Topology

- Add one router (Cisco 2911) and one switch (Cisco 2960).
- Add five PCs and one network printer.
- Connect the router to the switch using a copper straight-through cable.
- Connect all five PCs and the printer to the switch using copper straight-through cables.

Step 2: Configure the Router

Enter the following commands in the router CLI:

enable

configure terminal

interface gigabitEthernet 0/0

ip address 192.168.1.254 255.255.255.0

no shutdown

exit

ip dhcp excluded-address 192.168.1.254 192.168.1.100

ip dhcp pool OFFICE-LAN

network 192.168.1.0 255.255.255.0

default-router 192.168.1.254

dns-server 8.8.8.8

exit

end

write memory

Important: The excluded-address command above reserves 192.168.1.254 and 192.168.1.100 plus the entire range between them. To keep the configuration simple and reserve only the router address, use this command instead:

ip dhcp excluded-address 192.168.1.254

If you already configured the broader exclusion, remove it with no ip dhcp excluded-address 192.168.1.254 192.168.1.100 before applying the single-address exclusion.

Step 3: Assign IP Addresses

Use the Class C private network 192.168.1.0/24.

Device	IP address	Subnet mask	Default gateway
Router	192.168.1.254	255.255.255.0	—
PC0	DHCP assigned	255.255.255.0	192.168.1.254
PC1	DHCP assigned	255.255.255.0	192.168.1.254
PC2	DHCP assigned	255.255.255.0	192.168.1.254
PC3	DHCP assigned	255.255.255.0	192.168.1.254
PC4	DHCP assigned	255.255.255.0	192.168.1.254
Printer0	DHCP assigned or static	255.255.255.0	192.168.1.254

On each PC, open Desktop → IP Configuration → DHCP. For the printer, open its Config tab and select DHCP, as shown in your screenshot.

Step 4: Verify Connectivity

On each PC, open Desktop → Command Prompt and run:

ipconfig

ping 192.168.1.254

Then ping the IP address assigned to another PC and the printer. For example:

ping 192.168.1.101

Replace the example address with the actual IP address shown on the target device.

You can also verify DHCP leases on the router:

show ip dhcp binding

show ip interface brief

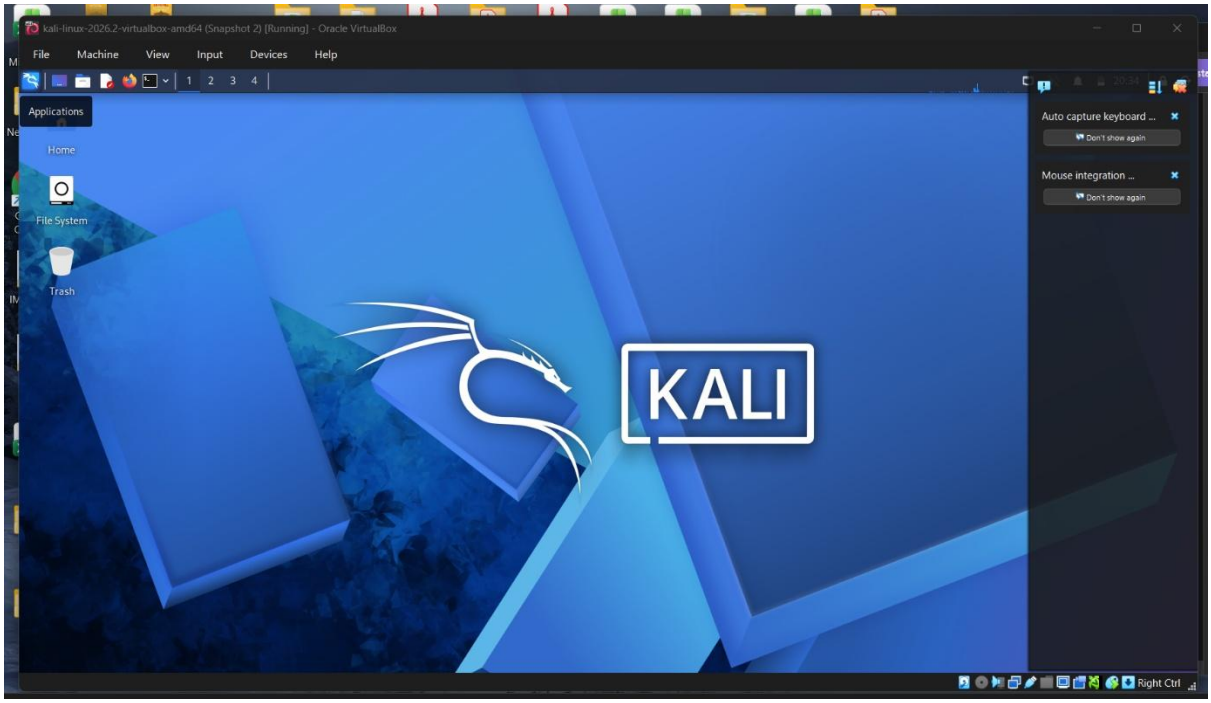
Step 5: Document the Result

Your existing screenshot shows the topology and a successful ping to the router. Include it as evidence, but take additional screenshots showing:

1. The complete topology with all seven devices.
2. The router DHCP configuration.
3. The IP configuration of each device, or a completed IP addressing table.
4. Successful pings between devices, including the printer.

Result: The LAN topology is built. The router ping is successful; verify PC-to-PC and PC-to-printer pings before claiming that all devices are reachable.

Task 4: Clean OS Installation



To perform a clean installation of Linux using an ISO file, manually partition the virtual disk, optimize startup programs and services, and verify that the operating system boots without errors.

Step 1: Prepare the Virtual Machine

1. Open Oracle VirtualBox.
2. Create a new virtual machine and select Linux as the operating system type.
3. Attach the Kali Linux ISO file to the virtual optical drive.
4. Allocate sufficient RAM, CPU cores, and virtual disk space.
5. Start the virtual machine and select the graphical installation option.

Step 2: Perform a Clean Installation

1. Select the language, location, and keyboard layout.
2. Configure the hostname and create a user account.
3. When prompted for disk partitioning, select the manual partitioning option.
4. Select the virtual disk intended for the installation. Ensure that no important data will be erased.

Step 3: Manually Partition the Disk

For a virtual disk of approximately 40 GB, an example partition layout is:

Partition	Size	Purpose

EFI System Partition	512 MB	Boot files for UEFI systems
Root (/)	30 GB	Operating system and applications
Swap	4 GB	Additional memory support
Remaining space	About 5.5 GB	Available for future use

The EFI partition is required for a UEFI installation. For a BIOS-based installation, the partition layout may differ. Choose the correct partition types and mount points, then confirm the changes before formatting the virtual disk.

Step 4: Complete Installation

1. Select the required desktop environment and system utilities.
2. Install the bootloader to the appropriate virtual disk.
3. Finish the installation and restart the virtual machine.
4. Detach the ISO or move the virtual disk to the first boot position if necessary.
5. Log in to Kali Linux and verify that the desktop loads correctly.

Step 5: Optimize Startup Programs and Services

Open the Kali Linux terminal and inspect the boot performance:

systemd-analyze

systemd-analyze blame

systemctl --type=service --state=running

- Identify services that consume significant startup time.
- Review unnecessary startup applications in the desktop's session and startup settings.
- Disable a service only after confirming that it is not required. For example, if the Bluetooth service is unnecessary on this workstation:

sudo systemctl disable bluetooth.service

Do not disable essential networking, security, display, or system services.

Step 6: Verify the Installation

Run the following commands:

lsblk

df -h

systemctl --failed

- lsblk verifies the disk and partition layout.
- df -h displays filesystem usage.

- systemctl --failed checks for failed system services.

Restart the virtual machine and confirm that Kali Linux boots successfully without installation or startup errors.

Result

The clean installation is successful when Kali Linux boots normally, the manually created partitions are present, and unnecessary startup programs or services have been reviewed and optimized.

Section C

Mini Project: Nexus Innovations — Diagnosing Random System Reboots

1. Three Hardware Components Most Likely to Cause Random Reboots

1. CPU (Processor)

The CPU generates significant heat during heavy rendering tasks. If the cooling system is inadequate, the processor may overheat and cause the system to shut down or restart to protect itself.

2. PSU (Power Supply Unit)

The PSU supplies electrical power to the CPU, GPU, motherboard, and other components. If it cannot provide sufficient power during heavy workloads, the workstation may restart unexpectedly.

3. GPU (Graphics Processing Unit)

Rendering tasks place a heavy load on the GPU, increasing its power consumption and temperature. Overheating or unstable GPU power delivery can cause system instability and unexpected restarts.

2. Why Systems Pass POST in the Morning but Fail in the Afternoon

In the morning, the workstations are relatively cool, and the components can operate normally during startup. POST checks whether essential hardware is functioning correctly, but it does not test the system under sustained heavy workloads.

During the afternoon, rendering applications increase CPU and GPU usage, producing more heat and consuming more power. Poor ventilation inside the desk cabinets traps hot air, causing temperatures to rise. The systems may then become unstable and restart because of overheating, insufficient power delivery, or hardware protection mechanisms.

3. Using a Stress-Test Tool to Confirm Thermal or Power Issues

I would use a trusted monitoring and stress-testing tool, such as OCCT, to test the workstations.

1. Check the CPU and GPU temperatures at idle using a hardware monitoring tool.
2. Run a CPU stress test and monitor temperatures, clock speeds, and system stability.
3. Run a GPU stress test and observe GPU temperatures and power consumption.
4. If temperatures remain within safe limits, perform a combined CPU and GPU stress test to increase total system power demand.
5. Check whether the workstation restarts, overheats, or reports hardware errors during testing.
6. Review system logs, including Windows Event Viewer, for unexpected shutdowns or hardware-related errors.

Interpretation: If temperatures rise excessively before a restart, overheating is a strong possibility. If the system restarts under combined load despite safe temperatures, investigate the PSU, power connections, and electrical supply. A stress test can provide evidence, but further testing may be needed to confirm the exact cause.

4. Three Specific Steps to Remediate Frequent Restarts

Step 1: Improve Cooling and Ventilation

- Move the workstations out of poorly ventilated cabinets or provide adequate airflow.
- Clean dust from fans, heatsinks, and air vents.
- Check that CPU and GPU cooling fans operate correctly.
- Replace thermal paste if necessary and ensure heatsinks are mounted correctly.

Step 2: Check the PSU and Power Connections

- Verify that the PSU has sufficient rated wattage and suitable power connectors for the CPU and GPU.
- Inspect power cables and connectors for loose connections or damage.
- Test with a known-good, adequately rated PSU if power delivery is suspected.
- Use a suitable UPS if the electrical supply is unstable.

Step 3: Monitor Temperatures and Test System Stability

- Monitor CPU and GPU temperatures during rendering.
- Update relevant BIOS/UEFI settings and hardware drivers when appropriate.
- Run memory diagnostics and review system logs if the problem continues.
- Repeat stress tests after repairs to verify that the workstations remain stable under load.

5. Technical Justification for a PSU Upgrade

A PSU upgrade may be necessary if the existing power supply cannot provide sufficient stable power for the CPU and GPU during heavy rendering workloads. An overloaded, faulty, or degraded PSU may cause sudden restarts.

However, a PSU upgrade is not automatically required. If monitoring confirms overheating due to poor ventilation or inadequate cooling, improving airflow and repairing the cooling system may resolve the issue without replacing the PSU. The PSU should be upgraded only if its capacity, condition, or power delivery is inadequate for the workstation's hardware.

Result

The most likely causes of the random restarts are CPU overheating, GPU overheating, and PSU-related power instability. Improving ventilation, checking cooling components, testing power delivery, and performing controlled stress tests can help identify the root cause and restore workstation stability.

Section D

STEP 1 · BUILD WITH AI

Build with AI — IP Addressing and Subnetting Plan

1. Exact Prompt Used

The following prompt was entered into ChatGPT to generate the IP addressing plan.

Generate an IP addressing and subnetting plan for Nexus Innovations' startup office network containing 5 computers (PC0, PC1, PC2, PC3, PC4) and 1 network printer (Printer0). Use a private Class C IPv4 network. Specify the network address, subnet mask, CIDR notation, usable host range, broadcast address, and default gateway. Assign a unique IP address to each computer and the printer. Also identify and reserve suitable IP addresses for future expansion. Present the result in a clear table.

Copy prompt

2. AI-Generated IP Addressing Plan

The AI-generated plan uses the private Class C network 192.168.1.0/24.

Network detail	Value
Network address	192.168.1.0
Subnet mask	255.255.255.0
CIDR notation	/24
Usable host range	192.168.1.1 – 192.168.1.254
Broadcast address	192.168.1.255
Default gateway	192.168.1.254

3. IP Address Allocation

Device name	Assigned IP address	Purpose
Router0	192.168.1.254	Default gateway
PC0	192.168.1.10	Employee workstation
PC1	192.168.1.11	Employee workstation
PC2	192.168.1.12	Employee workstation
PC3	192.168.1.13	Employee workstation
PC4	192.168.1.14	Employee workstation
Printer0	192.168.1.20	Network printer

4. Addresses Reserved for Future Expansion

The following addresses are reserved for future devices:

Reserved range	Purpose
192.168.1.21 – 192.168.1.49	Additional printers and workstations
192.168.1.50 – 192.168.1.99	Servers and network devices
192.168.1.100 – 192.168.1.199	Future employee computers and other devices

These addresses should be excluded from the DHCP allocation pool if they are intended for static assignments or future reservations. The router address 192.168.1.254 must also be excluded from DHCP.

STEP 2 · TEST & DEBUG (WITHOUT AI)

1. Verify the Subnet Details

Item	Verification result
Network address	192.168.1.0 — correct
Subnet mask	255.255.255.0 — correct
CIDR notation	/24 — correct
Usable host range	192.168.1.1 – 192.168.1.254 — correct
Broadcast address	192.168.1.255 — correct
Total usable host addresses	254 — sufficient for the network

A /24 subnet provides 256 total addresses, of which 254 are usable by hosts because the network and broadcast addresses cannot be assigned to devices.

2. Verify IP Address Allocation

Device	IP address	Status
Router0	192.168.1.254	Valid
PC0	192.168.1.10	Valid
PC1	192.168.1.11	Valid
PC2	192.168.1.12	Valid
PC3	192.168.1.13	Valid
PC4	192.168.1.14	Valid
Printer0	192.168.1.20	Valid

All assigned addresses are within the usable host range. There are no duplicate IP addresses or overlapping subnet ranges in this allocation.

3. Verify Future Expansion Reservations

The proposed expansion ranges are:

- 192.168.1.21 – 192.168.1.49
- 192.168.1.50 – 192.168.1.99
- 192.168.1.100 – 192.168.1.199

These ranges are consecutive and non-overlapping. They provide space for additional printers, servers, and workstations.

However, the ranges must be excluded from DHCP or managed through appropriate DHCP reservations to prevent future conflicts with statically assigned addresses.

4. Corrected IP Addressing Plan

No subnet calculation or device IP address needs to be changed because the original plan is valid. The correction is to clarify how the reserved addresses will be managed.

Network detail	**Corrected plan**
Network	192.168.1.0/24
Subnet mask	255.255.255.0
Network address	192.168.1.0
Broadcast address	192.168.1.255
Default gateway	192.168.1.254
Device addresses	192.168.1.10–14 and 192.168.1.20
Future expansion	192.168.1.21–199, divided into the ranges above
DHCP management	Exclude static addresses and reserve addresses where appropriate

5. Verification Note (3–4 Lines)

I checked the network address, subnet mask, usable host range, and broadcast address and found them correct. All seven assigned device addresses are unique and fall within the usable host range. The future expansion ranges do not overlap, but they must be managed correctly to avoid DHCP conflicts. Therefore, I retained the original IP addresses and clarified the DHCP reservation and exclusion requirements.